

Mechanics of Many-Particle Systems

Classical Mechanics– Lecture Notes

1 Conservation of Linear Momentum

We extend the single-particle results to a collection of N particles labelled $\alpha = 1, 2, \dots, N$. Each particle carries mass m_α and has position vector $\vec{r}_\alpha$ measured from a fixed origin O . The total mass of the system is

$$M = \sum_{\alpha=1}^N m_\alpha. \quad (1)$$

The total linear momentum is the vector sum of the individual momenta:

$$\vec{P} = \sum_{\alpha=1}^N \vec{p}_\alpha = \sum_{\alpha=1}^N m_\alpha \vec{v}_\alpha. \quad (2)$$

The net force acting on particle α consists of an external contribution $\vec{F}_\alpha^{\text{ext}}$ together with the internal forces $\vec{F}_{\alpha\beta}$ exerted by every other particle β :

$$\vec{F}_\alpha = \vec{F}_\alpha^{\text{ext}} + \sum_{\substack{\beta=1 \\ \beta \neq \alpha}}^N \vec{F}_{\alpha\beta}. \quad (3)$$

(The term $\vec{F}_{\alpha\alpha}$ is absent because a particle cannot exert a force on itself.) Differentiating the total momentum with respect to time therefore yields

$$\dot{\vec{P}} = \sum_{\alpha=1}^N \dot{\vec{p}}_\alpha = \sum_{\alpha=1}^N \vec{F}_\alpha = \sum_{\alpha=1}^N \vec{F}_\alpha^{\text{ext}} + \sum_{\alpha=1}^N \sum_{\substack{\beta=1 \\ \beta \neq \alpha}}^N \vec{F}_{\alpha\beta}. \quad (4)$$

Newton's third law states that the force exerted by β on α is equal and opposite to the force exerted by α on β :

$$\vec{F}_{\alpha\beta} = -\vec{F}_{\beta\alpha}. \quad (5)$$

Consequently the double sum of internal forces can be rearranged by pairing terms:

$$\sum_{\alpha} \sum_{\beta \neq \alpha} \vec{F}_{\alpha\beta} = \frac{1}{2} \sum_{\alpha} \sum_{\beta \neq \alpha} (\vec{F}_{\alpha\beta} + \vec{F}_{\beta\alpha}) = \frac{1}{2} \sum_{\alpha} \sum_{\beta \neq \alpha} (\vec{F}_{\alpha\beta} - \vec{F}_{\alpha\beta}) = \vec{0}. \quad (6)$$

Every internal force cancels its partner, and equation (4) reduces to

$$\dot{\vec{P}} = \vec{F}^{\text{ext}}, \quad (7)$$

where the net external force is defined by

$$\vec{F}^{\text{ext}} = \sum_{\alpha=1}^N \vec{F}_\alpha^{\text{ext}}. \quad (8)$$

If the net external force vanishes, the total linear momentum of the system is a constant of the motion:

$$\vec{P} = \sum_{\alpha=1}^N m_{\alpha} \vec{v}_{\alpha} = \text{constant}. \quad (9)$$

2 Centre of Mass

The centre-of-mass (CM) position vector is the mass-weighted average of the individual position vectors:

$$\vec{R} = \frac{1}{M} \sum_{\alpha=1}^N m_{\alpha} \vec{r}_{\alpha}. \quad (10)$$

In Cartesian components this definition reads

$$\begin{aligned} X &= \frac{1}{M} \sum_{\alpha=1}^N m_{\alpha} x_{\alpha}, \\ Y &= \frac{1}{M} \sum_{\alpha=1}^N m_{\alpha} y_{\alpha}, \\ Z &= \frac{1}{M} \sum_{\alpha=1}^N m_{\alpha} z_{\alpha}. \end{aligned} \quad (11)$$

Differentiating (10) with respect to time gives the centre-of-mass velocity:

$$\dot{\vec{R}} = \frac{1}{M} \sum_{\alpha=1}^N m_{\alpha} \dot{\vec{r}}_{\alpha} = \frac{1}{M} \sum_{\alpha=1}^N m_{\alpha} \vec{v}_{\alpha} = \frac{\vec{P}}{M}. \quad (12)$$

Rearrangement immediately produces the important identity

$$\vec{P} = M \dot{\vec{R}}. \quad (13)$$

Thus the total linear momentum of the N -particle system is identical to the momentum of a single fictitious particle of mass M that moves with the centre-of-mass velocity.

A second differentiation, combined with the momentum equation (7), yields the equation of motion of the centre of mass:

$$M \ddot{\vec{R}} = \vec{F}^{\text{ext}}. \quad (14)$$

The centre of mass therefore accelerates exactly as though all the external forces acted at a single point of mass M . This is the fundamental reason we may treat an extended body as a point particle when only its translational motion is of interest.

3 Conservation of Angular Momentum

The total angular momentum of an N -particle system relative to a fixed origin O is defined by

$$\vec{L} = \sum_{\alpha=1}^N \vec{r}_{\alpha} \times \vec{p}_{\alpha} = \sum_{\alpha=1}^N \vec{r}_{\alpha} \times m_{\alpha} \vec{v}_{\alpha}. \quad (15)$$

3.1 Time derivative of total angular momentum

Differentiate term by term with respect to time:

$$\dot{\vec{L}} = \sum_{\alpha=1}^N \frac{d}{dt} (\vec{r}_{\alpha} \times \vec{p}_{\alpha}) = \sum_{\alpha=1}^N \left(\dot{\vec{r}}_{\alpha} \times \vec{p}_{\alpha} + \vec{r}_{\alpha} \times \dot{\vec{p}}_{\alpha} \right). \quad (16)$$

The first cross product vanishes identically:

$$\dot{\vec{r}}_{\alpha} \times \vec{p}_{\alpha} = \vec{v}_{\alpha} \times (m_{\alpha} \vec{v}_{\alpha}) = m_{\alpha} (\vec{v}_{\alpha} \times \vec{v}_{\alpha}) = \vec{0}. \quad (17)$$

Newton's second law supplies $\dot{\vec{p}}_{\alpha} = \vec{F}_{\alpha}$, so

$$\dot{\vec{L}} = \sum_{\alpha=1}^N \vec{r}_{\alpha} \times \vec{F}_{\alpha}. \quad (18)$$

The right-hand side is precisely the total torque on the system about the origin O .

3.2 Separation into external and internal torques

Write the force on particle α as the sum of the external force and all internal forces:

$$\vec{F}_{\alpha} = \vec{F}_{\alpha}^{\text{ext}} + \sum_{\beta \neq \alpha} \vec{F}_{\alpha\beta}. \quad (19)$$

Substitution into (18) yields two distinct contributions:

$$\dot{\vec{L}} = \underbrace{\sum_{\alpha} \vec{r}_{\alpha} \times \vec{F}_{\alpha}^{\text{ext}}}_{\vec{\Gamma}^{\text{ext}}} + \sum_{\alpha} \sum_{\beta \neq \alpha} \vec{r}_{\alpha} \times \vec{F}_{\alpha\beta}. \quad (20)$$

The first sum is the net external torque $\vec{\Gamma}^{\text{ext}}$. We now show that the double sum of internal torques vanishes when the internal forces are central and obey Newton's third law.

3.3 Cancellation of internal torques

Consider the double sum

$$S = \sum_{\alpha=1}^N \sum_{\substack{\beta=1 \\ \beta \neq \alpha}}^N \vec{r}_{\alpha} \times \vec{F}_{\alpha\beta}. \quad (21)$$

Because the indices α and β run over the same set, we may also write

$$S = \sum_{\alpha} \sum_{\beta \neq \alpha} \vec{r}_{\beta} \times \vec{F}_{\beta\alpha}. \quad (22)$$

Adding the two expressions gives

$$2S = \sum_{\alpha} \sum_{\beta \neq \alpha} \left(\vec{r}_{\alpha} \times \vec{F}_{\alpha\beta} + \vec{r}_{\beta} \times \vec{F}_{\beta\alpha} \right). \quad (23)$$

Newton's third law asserts that $\vec{F}_{\beta\alpha} = -\vec{F}_{\alpha\beta}$, so

$$2S = \sum_{\alpha} \sum_{\beta \neq \alpha} \left(\vec{r}_{\alpha} \times \vec{F}_{\alpha\beta} - \vec{r}_{\beta} \times \vec{F}_{\alpha\beta} \right) = \sum_{\alpha} \sum_{\beta \neq \alpha} (\vec{r}_{\alpha} - \vec{r}_{\beta}) \times \vec{F}_{\alpha\beta}. \quad (24)$$

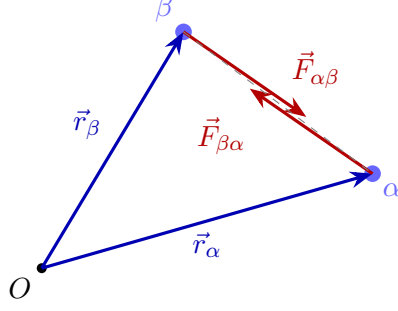


Figure 1: Internal forces that are equal, opposite and collinear produce equal and opposite torques that cancel.

The vector $\vec{r}_\alpha - \vec{r}_\beta$ points from particle β to particle α . If the internal force $\vec{F}_{\alpha\beta}$ is *central*, it lies exactly along this line:

$$\vec{F}_{\alpha\beta} \parallel (\vec{r}_\alpha - \vec{r}_\beta). \quad (25)$$

The cross product of two parallel vectors is zero, and therefore every term in the double sum vanishes:

$$S = 0. \quad (26)$$

Consequently the internal torques cancel completely and we obtain the compact result

$$\dot{\vec{L}} = \vec{\Gamma}^{\text{ext}} = \sum_{\alpha=1}^N \vec{r}_\alpha \times \vec{F}_\alpha^{\text{ext}}. \quad (27)$$

3.4 Conservation law

Whenever the net external torque about the chosen origin vanishes,

$$\vec{\Gamma}^{\text{ext}} = \vec{0}, \quad (28)$$

the total angular momentum is a constant of the motion:

$$\vec{L} = \sum_{\alpha} \vec{r}_\alpha \times \vec{p}_\alpha = \text{constant}. \quad (29)$$

3.5 Decomposition relative to the centre of mass

It is often useful to separate the total angular momentum into a contribution that comes from the motion of the centre of mass and a contribution that comes from motion relative to the centre of mass. Write

$$\vec{r}_\alpha = \vec{R} + \vec{r}'_\alpha, \quad (30)$$

where the relative vectors satisfy the centre-of-mass condition

$$\begin{aligned} \sum_{\alpha=1}^N m_\alpha \vec{r}'_\alpha &= \sum_{\alpha=1}^N m_\alpha (\vec{r}_\alpha - \vec{R}) \\ &= \sum_{\alpha=1}^N m_\alpha \vec{r}_\alpha - M \vec{R} \\ &= M \vec{R} - M \vec{R} \\ &= 0 \end{aligned} \quad (31)$$

Differentiating (30) gives the corresponding velocity decomposition

$$\vec{v}_\alpha = \dot{\vec{R}} + \dot{\vec{r}}'_\alpha. \quad (32)$$

Substitute both expressions into the definition of total angular momentum:

$$\begin{aligned} \vec{L} &= \sum_\alpha (\vec{r}_\alpha \times \vec{p}_\alpha) \\ &= \sum_\alpha (\vec{r}_\alpha \times m_\alpha \vec{v}_\alpha) \\ &= \sum_\alpha (\vec{R} + \vec{r}'_\alpha) \times m_\alpha (\dot{\vec{R}} + \dot{\vec{r}}'_\alpha) \\ &= \sum_\alpha \vec{R} \times m_\alpha \dot{\vec{R}} + \sum_\alpha \vec{R} \times m_\alpha \dot{\vec{r}}'_\alpha + \sum_\alpha \vec{r}'_\alpha \times m_\alpha \dot{\vec{R}} + \sum_\alpha \vec{r}'_\alpha \times m_\alpha \dot{\vec{r}}'_\alpha. \end{aligned} \quad (33)$$

Examine each of the four sums in turn.

1) The first sum factors immediately:

$$\sum_\alpha \vec{R} \times m_\alpha \dot{\vec{R}} = \vec{R} \times \left(\sum_\alpha m_\alpha \right) \dot{\vec{R}} = \vec{R} \times M \dot{\vec{R}} = \vec{R} \times \vec{P}. \quad (34)$$

2) The second sum vanishes by the centre-of-mass condition. Differentiating (31) with respect to time yields

$$\sum_\alpha m_\alpha \dot{\vec{r}}'_\alpha = \frac{d}{dt} \sum_\alpha m_\alpha \vec{r}'_\alpha = \vec{0}, \quad (35)$$

and therefore

$$\sum_\alpha \vec{R} \times m_\alpha \dot{\vec{r}}'_\alpha = \vec{R} \times \left(\sum_\alpha m_\alpha \dot{\vec{r}}'_\alpha \right) = \vec{0}. \quad (36)$$

3) The third sum likewise vanishes:

$$\sum_\alpha \vec{r}'_\alpha \times m_\alpha \dot{\vec{R}} = \left(\sum_\alpha m_\alpha \vec{r}'_\alpha \right) \times \dot{\vec{R}} = \vec{0} \times \dot{\vec{R}} = \vec{0}. \quad (37)$$

4) The fourth sum remains as the intrinsic angular momentum relative to the centre of mass:

$$\sum_\alpha \vec{r}'_\alpha \times m_\alpha \dot{\vec{r}}'_\alpha. \quad (38)$$

Collecting the surviving terms we obtain the clean decomposition

$$\vec{L} = \vec{R} \times \vec{P} + \sum_{\alpha=1}^N \vec{r}'_\alpha \times m_\alpha \dot{\vec{r}}'_\alpha. \quad (39)$$

The first term is the angular momentum of a single particle of mass M located at the centre of mass; the second term is the angular momentum of the internal motion measured relative to the centre of mass.